# Tony Jie Wang

Mobile: +1 (267)-403-8674 | E-mail: tonyw3@seas.upenn.edu | LinkedIn: Jie Wang's LinkedIn
Website: everloom-129.github.io | GitHub: github.com/Everloom-129 | Twitter: twitter.com/JieWang_ZJUI

## EDUCATION

**University of Pennsylvania** *(GPA: 3.63/4.0)* — Philadelphia, PA
M.S.E Robotics, Advisor: Prof. Dinesh Jayaraman & Prof. Kostas Daniilidis — Aug 2024 – Jun 2026

**Zhejiang University** *(GPA: 3.83/4.0)* — Zhejiang, China
BEng. Electronic and Computer Engineering, Advisor: Prof. Said Mikki — Aug 2020 – Jun 2024

**University of Illinois at Urbana-Champaign** *(GPA: 3.50/4.0)* — Urbana, IL
BSc. Computer Engineering, Advisor: Prof. Deepak Vasisht — Aug 2020 – Jun 2024

## INDUSTRY EXPERIENCE

**DynaRobotics**
*DYNA-2 Research Team, Mentor: Dr. Jason Ma* — *Redwood City, CA*
Research Intern — Jun 2025 – Aug 2025

- Prototyped **Agentic Reasoning** project**:** designed an autoregressive head to let VLAs talk: interact with multiple users, verify current progress and maintain task memory using one model. [video]
- Engineered large-scale data pipelines by curating, filtering, and normalizing external datasets into pre-training recipes.
- Developed a subtask prediction module for task progress estimation, achieving 90% accuracy on Napkin Folding.
- Deployed commercial models for customer tasks including *Napkin Folding, Cloth Stacking* and *Candy Pick-and-Place* .

**International Digital Economy Academy**
*Computer Vision & Robotics Department, Mentor: Dr. Xiaoke Jiang* — *Shenzhen, China*
MLE Intern — Jun 2024 – Sep 2024

- Diagnosed I/O bottlenecks between SLURM clusters, accelerated data loading throughput by 16.7%
- Adapted Grounding DINO 1.6 as a vision encoder to train VLMs on Motion-X++, large-scale human motion dataset.
- Trained, Finetuned and Optimized Object-Detection VLM for human behavior recognition.
- Contributed to the DINO-X, a SOTA vision foundation model family for open-world understanding. [model]

## PUBLICATIONS

- **Wang, J.** , Rahman, F., Pasumarti, V., Hu, E. S., Krisha, A., Gu, Jiatao., Daniilidis, K., Jayaraman, D. *AttnToVLA: An Attention Mechanism Perspective on VLAs* Manuscript.
- Song, Y.*, Le, L.*, Park, Y., **Wang, J.**, Shi, J., Liu, L., Gu, J., Eaton, E., Jayaraman, D., Daniilidis, K. *OmniGuide: Universal Guidance Fields for Enhancing Generalist Robot Policies.* Under review.
- Shen, W.*, Kumar, N.*, Chintalapudi, S.*, **Wang, J.,** Watson, C., Hu, E. S., Cao, J., Jayaraman, D., Kaelbling, L. P., Lozano-Pérez,T. *TiPToP: A Modular Open-Vocabulary Planning System for Robotic Manipulation*. Under review.
- Hu, E. S.* **Wang, J.***, Yuan, X.* Luo, F. Li, M. Lambrechts, G. Rybkin, O. Jayaraman, D. (2025) *AAWR: Real-World Reinforcement Learning of Interactive Perception Behaviors,* (NeurIPS2025 & ARLET Workshop)
- **Wang, J.*** Leonard, M. Daniilidis, K. Jayaraman, D. Hu, E. S.(2025) *Evaluating pi0 in the Wild: Strengths, Problems, and the Future of Generalist Robot Policies*, GRASP Lab Blog.
- RoboArena Team. (2025) *RoboArena: Distributed Real-World Evaluation of Generalist Robot Policies*, **CoRL 2025 Oral**
- Shi, J. Zhao, Z. Wang. T. Pedroza, I. Luo, A. **Wang, J.** Ma, J. Jayaraman. D (2025) *ZeroMimic: Distilling Robotic Manipulation Skills from Web Videos,* . **ICRA 2025** & Best Paper at **CVPR 2025** 3DVLM Workshop**.**
- Zhu, X., Li, Z., Jiang, Y., Xu, J., **Wang, J.**, & Bai, X. *"Real-time V2V Communication Network Cooperative Control System through Distributed Database,"* **oral** at ICICT2024, London, UK.
- Xiang, X., Lei, Z., **Wang, J.** Zheng, Q., Huang Y. *"Visionary Co-Driver: LLMs Enhance Driver Risk Perception with ARHUD",* **IEEE Transactions on Intelligent Transportation Systems** (IEEE-ITS)

## Research Experience

**2025 Master Thesis: Explaining and Enhancing VLAs at Attention Level** — GRASP Lab
*Keywords: Mechanistic Interpretability, VLAs* — *Prof. Kostas Daniilidis*

- Benchmarked Pi05, Gr00t N1.6, SmolVLA performance on complex spatial reasoning tasks.
- Visualized the attention map of VLAs to study the correlation between text, image and action.
- Offline evaluated DROID dataset using attention-object bbox correlation as intermediate metrics.
- Proposed an attention-level guidance method for test-time training, improving VLAs capability under external camera view.

**2025 Evaluation of Generalist Policies in the Wild** — GRASP Lab
*Keywords: Model Evaluation, Foundation Models, VLA*

- Led comprehensive observation study on pi0-fast-droid models across 300+ real-world tasks. [blog]
- Conducted A/B Comparison of evaluation pool and distributed scoring system.
- Co-organized the RoboArena Workshop at CoRL 2025
- Collaboration: TipTop with MIT on TAMP v.s. End2End methods for tabletop pick-and-place tasks. [Result]
- ***Ongoing collaboration:*** *RoboArena-Sim with Lightwheel Inc. for large-scale real2sim evaluations.*

**2025 AAWR: Real-World Reinforcement Learning of Active Perception Behaviors** GRASP Lab
*Keywords: Active Perception, RL, VLA* *Prof. Dinesh Jayaraman*

- Proposed **Asymmetric-AWR**, a novel RL framework that leverages Vision Foundation Models as privileged visual critics during training to scaffold active perception policies, significantly outperforming BC and standard AWR baselines.
- Built hardware and perception stack for Koch Arm, blind pick and place got 89% success rate using online RL.
- Refactored DROID infra as **EVA**, supporting GELLO and 3DSpaceMouse interfaces, accelerating data collection speed by ~20s per trial with better control compared to VR Controller.
- AAWR outperform all baselines in VLA Active Perception in Franka Robot Arm with only 50 demonstrations.

**2024 Zero-shot Manipulation using VLMs as Planner** GRASP Lab
*Keywords: Foundation Models, Planning, Hybrid Control* *Prof. Dinesh Jayraman*

- Reproduced ReKep on a Mobile Franka Arm Platform with DROID, conducted experiment for ZeroMimic as a baseline.
- Reproduced SPHINX for hybrid salient point control using transformer and diffusion policy.
- Built a video processing tool as shared infra, upgraded the physical lighting system for the lab.

**2023 Visionary Co-Driver: LLMs Enhance Driver Risk Perception with ARHUD** Zhejiang University
*Keywords: Autonomous Driving, Foundation Models, HCI* *Advisor: Prof. Wei Xiang*

- Engineered an LLM-driven risk assessment framework for human-machine co-driving; integrated the system to interpret monocular video streams and generate eye-tracking, context-aware risk warnings via AR-HUD.
- Designed the Human-Computer Interface incorporating eye-tracking to monitor driver attention
- Conducted user study (N=41), showing 2x improved risk awareness and 0.5x reaction times.

## PROJECT EXPERIENCE

**2024 Undergrad Senior Design: Intelligent Pour-Over Coffee Machine** *[Thesis]* Haining, China

- Built a pour-over coffee machine by reverse engineering.
- Guaranteed food-grade quality coffee for drink.
- Customizable coffee recipe according to user preferences.

**2023 CS438: Communication Networks Wireless Project** *[Report]* Urbana, IL

- Developed an open-source Python tool for comprehensive wireless network analysis, focusing on Wi-Fi access points roaming mechanisms and signal strength heatmap generation.
- Designed a procedure-oriented data pipeline architecture including coordinate construction, data collection, data preprocessing, heatmap generation, and individual AP analysis.

**2021 Sim & Real Experiment on Baidu Apollo Autonomous Vehicle** Haining, China

- Reproduced Apollo7.0 simulation experiments on CARLA, testing across 10 maps.
- Assemble a Baidu Apollo D-kit Test Autonomous Vehicle (2/3 of real vehicle size).
- Constructed the high-solution LiDAR map of ZJU international campus via Baidu Apollo D-kit Vehicle.

## ACADEMIC CONTEST

**2023 Shell Eco-marathon Autonomous Programming Competition** *ROS, Carla, Python,*

- Developed path planning, perception, and control modules for simulation autonomous vehicles using the Robot Operating System (ROS) stack provided by the competition.
- Utilized the CARLA simulator with the Unreal Engine to test our vehicle in a simulated environment, with the goal of achieving the most efficient path planning according to the competition's ranking criteria.

**2022 International Mathematical Contest in Modeling (Honorable Mention)** *MATLAB, Python*

- Addressed the issue of water scarcity in the Colorado River in the United States by constructing a mathematical model for water-to-electricity supply using dynamic programming and Monte Carlo algorithm.
- Used SPSS time series analysis tools to predict water demand for each state, generated a demand matrix for the water system. By sensitivity analysis, demonstrated strategies for addressing conditions such as rapid depletion of water resources, involvement of renewable energy technologies, and application of conservation measures.

**2021 DJI RoboMaster University Robotics Competition** *Python, PyTorch, C/C++*

- Assisted in developing and optimizing vision algorithms for object detection via YOLOv4, contributing to auto-aiming and shooting system for real-time target engagement, improving hit accuracy by 20%.
- Supported the integration of vision systems on Jetson Nano with embedded platforms on DJI manifold2, achieving a 30 FPS during live competition on the industrial camera.

## SERVICE

Reviewer for RSS2025, NeurIPS2025, ICLR2026
Teaching Assistant in CIS5800: Machine Perception 2026
GRASP Lab Student Faculty Industry (SFI) Committee Member at UPenn 2025
Member, Robotics Entrepreneur Club (PERC) at UPenn 2025
Teaching Assistant in MEAM5200: Introduction to Robotics 2025
Teaching Assistant in PHIL 206: Early Modern Philosophy 2024
Head Teaching Assistant in MATH213: Introduction to Discrete Mathematics 2023
Founder & President, PhiloCoffee Club at ZJU 2023